

Advanced Machine Learning Strategies for Predicting Therapy Response in Preclinical Glioblastoma using Longitudinal MRI

Javier González¹, Ana Paula Candiota^{2,3}, and Alfredo Vellido^{1,3,4*}

¹Department of Computer Science, Universitat Politècnica de Catalunya (UPC), Barcelona, Spain

²Department of Biochemistry and Molecular Biology, Universitat Autònoma de Barcelona (UAB), Cerdanyola del Vallès, Spain

³Centro de Investigación Biomédica en Red en Bioingeniería, Biomateriales y Nanomedicina (CIBER-BBN), Madrid, Spain

⁴Intelligent Data Science and Artificial Intelligence (IDEAI) Research Center, UPC, Barcelona, Spain

*Corresponding author: alfredo.vellido@upc.edu

Abstract

Glioblastoma (GB) is the most aggressive primary brain tumor, characterized by a poor prognosis, limited response to therapy, and high rates of recurrence. Early therapeutic response assessment is challenging due to phenomena such as pseudoresponse and pseudoprogression. This study explores the potential of advanced machine learning (ML) strategies to predict long-term therapy outcomes using longitudinal T2-weighted Magnetic Resonance Imaging (MRI) data from a preclinical GL261 glioblastoma mouse model, acquired prior and during treatment. We compare two distinct approaches: a classical pipeline based on radiomic features coupled with an XGBoost classifier, and a deep learning (DL) pipeline using a fine-tuned EfficientNetB0 model. Our results demonstrate that while the radiomics approach identifies interpretable imaging biomarkers and achieves good predictive performance ($AUC \approx 0.70$), the DL-based model outperforms it across all metrics, reaching an AUC of 0.85 and a sensitivity of 0.91. The DL model shows superior stability over time, particularly in early predictive windows, and better generalization across individual subjects. Interpretability analysis via Grad-CAM confirms that the DL model's predictions are driven by anatomically relevant features. These findings highlight the significant potential of DL, enhanced by transfer learning, to serve as a powerful non-invasive tool for the early prediction of treatment efficacy in glioblastoma, paving the way for more robust and personalized therapy monitoring.

1 Introduction

Glioblastoma (GB) stands as the most common and lethal primary brain tumor in adults, with a median survival of just 15-18 months despite aggressive multimodal treatment consisting of surgery, radiotherapy, and chemotherapy with temozolomide (TMZ), dating from more than two decades ago [1]. A major clinical challenge is the accurate and early assessment of treatment response. Standard evaluation criteria can be confounded by treatment-related effects such as pseudoprogression (transient, treatment-induced inflammation mimicking tumor growth) and pseudoresponse (transient decrease in the contrast enhancement area) [2], which complicates clinical decision-making in early scenarios.

Preclinical models, such as the murine GL261 glioblastoma model [3, 4], offer a controlled environment to study tumor evolution and response to therapy. Longitudinal Magnetic Resonance Imaging (MRI) in these models provides a rich source of data to develop and validate non-invasive biomarkers that could be later translated to clinical settings.

In recent years, Artificial Intelligence (AI) has shown great promise in medical image analysis. Two dominant paradigms have emerged: radiomics-based machine learning (ML) and deep learning (DL). Radiomics involves the high-throughput extraction of quantitative, handcrafted features from medical images, which describe tumor shape, intensity, and texture [5]. These features can provide interpretable insights but rely on domain-specific feature engineering. In contrast, DL, particularly with Convolutional Neural Networks (CNNs), learns relevant hierarchical features directly from raw image data, often achieving superior performance but at the cost of reduced interpretability [6–8]. Despite the rapid adoption of DL, radiomics remains attractive because handcrafted features offer greater interpretability and can provide biologically meaningful descriptors

of tumor morphology and heterogeneity. Consequently, direct comparisons between radiomics-based and DL-based frameworks remain important for understanding the trade-off between predictive performance and model interpretability.

This study aims to systematically compare these two approaches for the early prediction of long-term therapy outcomes (cure vs. relapse) in a preclinical GB model using longitudinal T2-weighted MRI, one of the workhorse sequences used in the MR-based scenario. We developed and evaluated both a classical radiomics-based pipeline and a modern DL-based pipeline, focusing on three key objectives:

1. To assess and compare the predictive performance of both methodologies.
2. To identify key imaging biomarkers associated with treatment response and relapse.
3. To explore the interpretability of both models to ensure potential clinical relevance.

By doing so, we seek to determine which of the studied strategies offers a more robust and scalable solution for developing non-invasive tools to improve glioblastoma therapy assessment.

In this context, the term *longitudinal* refers to the acquisition of multiple MRI examinations from the same animal at different time points throughout tumor evolution and treatment, rather than to a single cross-sectional acquisition. This design provides information about the temporal dynamics of tumor response. It is important to note, however, that although the data are longitudinal, they were not modeled as a multivariate time series in the sense of jointly classifying complete temporal sequences [9]. Instead, each imaging time point was treated as an individual observation, and temporal information was incorporated in the radiomics pipeline through engineered temporal descriptors (see Methods), while preserving the subject-level structure of the data during validation.

Related Work

ML and DL approaches have become increasingly important in neuro-oncology, particularly for the analysis of MRI data [10]. Traditional radiomics pipelines transform medical images into quantitative descriptors of tumor morphology, intensity, and texture, enabling predictive modeling using machine learning algorithms [5, 11]. Recent evidence continues to support the prognostic value of MRI-derived radiomic biomarkers in glioblastoma, although challenges related to reproducibility, feature stability, and external validation remain important barriers to clinical translation [12]. Despite the rapid adoption of DL methods, radiomics remains attractive because handcrafted features offer greater interpretability and can provide biologically meaningful descriptors of tumor morphology and heterogeneity. Consequently, direct comparisons between radiomics-based and DL-based frameworks remain important for understanding the trade-off between predictive performance and model interpretability.

More recently, DL methods have demonstrated the ability to learn hierarchical image representations directly from imaging data, reducing dependence on handcrafted feature engineering and achieving state-of-the-art performance in several neuroimaging tasks [6–8, 13]. Recent reviews have further highlighted the growing role of DL in glioblastoma imaging, where convolutional neural networks are increasingly capable of extracting clinically relevant information directly from MRI data without explicit feature engineering [14].

Most existing studies have focused on tumor detection, segmentation, or diagnostic classification. For example, Cinar and Yildirim proposed a hybrid ResNet50-based architecture for brain tumor detection from MRI, achieving high classification accuracy and demonstrating the effectiveness of convolutional neural networks for neuroimaging applications [15]. However, predicting therapeutic response remains considerably more challenging because it requires longitudinal analysis and the identification of subtle imaging changes that precede overt anatomical alterations.

Several studies have explored this problem in preclinical GB models. Núñez et al. combined radiomics, Convex-NMF decomposition, and feature-selection strategies to characterize treatment effects in GL261 glioblastoma using multimodal MRI and magnetic resonance spectroscopic imaging (MRSI) data, demonstrating the potential of quantitative imaging biomarkers for therapy assessment [16]. Ortega-Martorell et al. subsequently introduced one-dimensional convolutional neural networks for analyzing longitudinal spectroscopic data, achieving improved discrimination between responsive and non-responsive tumor regions and highlighting the capacity of DL to identify early therapy-related changes [17]. In parallel, Zhao et al. integrated MR-based metabolomics with molecular profiling of the tumor microenvironment, linking imaging biomarkers to immune-related mechanisms associated with tumor progression and relapse [18]. Prinzi et al. further demonstrated the value of explicitly modeling radiomic features as multivariate time series in dynamic contrast-enhanced breast MRI [9], an approach conceptually related to, but methodologically distinct from, the time-point-based strategy adopted here.

Although these studies have demonstrated the feasibility of imaging-based response monitoring, most rely on spectroscopic imaging, multimodal acquisitions, or highly specialized biomarkers. Comparatively fewer

investigations have examined whether standard T2-weighted MRI alone contains sufficient information to predict long-term therapeutic outcomes. The present study addresses this gap by directly comparing a radiomics-based ML framework with a transfer-learning DL approach for predicting relapse and cure from longitudinal T2-weighted MRI acquired in a preclinical GL261 glioblastoma model.

2 Results

2.1 Performance Comparison

Table 1 summarizes the performance of the best-performing radiomics and DL models under the Grouped Leave-One-Out (GLOO) evaluation framework.

The DL approach achieved the highest overall performance, obtaining an Area Under the Receiver Operating Characteristic Curve (AUC) of 0.8511 compared with 0.7015 for the radiomics pipeline. Improvements were observed across nearly all evaluation metrics, particularly sensitivity, where EfficientNet correctly identified 90.9% of cured animals compared with 51.5% for XGBoost.

Although the radiomics model achieved slightly higher specificity (0.8879 versus 0.7931), its substantially lower sensitivity indicates a tendency to classify uncertain cases as relapsing. In contrast, the DL model maintained a more favorable balance between sensitivity and specificity, resulting in better discrimination performance.

Model	Accuracy	AUC	Sensitivity	Specificity	PPV	NPV	F1w	F1m
Radiomics + XGBoost	0.8054	0.7015	0.5152	0.8879	0.5667	0.8655	0.8020	0.7081
EfficientNet FE + XGB	0.8054	0.8583	0.6212	0.8578	0.5541	0.8884	0.8092	0.7293
EfficientNet FT	0.8188	0.8511	0.9091	0.7931	0.5556	0.9684	0.8316	0.7808

Table 1: Performance comparison of radiomics and DL configurations. FE: frozen feature extractor; FT: fine-tuned. F1w and F1m denote the weighted and macro F1-scores, respectively. PPV and NPV denote the positive and negative predictive values, respectively.

2.2 Radiomics Model Analysis

Progressive refinement of the radiomics pipeline revealed the importance of rigorous feature selection in high-dimensional imaging datasets. Models trained using the complete feature set exhibited unstable behavior and reduced generalization, whereas statistical filtering and redundancy removal substantially improved predictive performance. A comparison of the resampling strategies evaluated for the XGBoost classifier is reported in Supplementary Figure S1.

The baseline configuration, trained on the full feature set, achieved high specificity (0.9494) and accuracy (0.8455) but poor sensitivity (0.3485), effectively defaulting to a relapse prediction under the strong class imbalance. Because correctly identifying cured animals was the primary objective, subsequent configurations targeted sensitivity while preserving specificity. Statistical filtering with the Mann–Whitney U test yielded the best overall discrimination (AUC 0.7015, sensitivity 0.5152); introducing lag-based temporal descriptors for the top-ranked features maintained this balance, and an additional variance-, correlation- and significance-based filtering step reached the highest precision (PPV 0.5926). These gains, although modest in absolute terms, were consistent and biologically coherent within a small, highly imbalanced cohort.

Feature importance analysis consistently identified a small subset of descriptors as dominant contributors to classification decisions, including Elongation, Dependence Entropy, Run Entropy, Short Run High Gray Level Emphasis, and High Gray Level Run Emphasis. These features collectively describe tumor morphology, structural organization, and texture heterogeneity.

Longitudinal analysis revealed distinct temporal trajectories between cured and relapsing tumors. In cured animals, Elongation (Figure 1) progressively decreased following treatment administration, whereas relapsing tumors maintained more irregular growth patterns throughout follow-up.

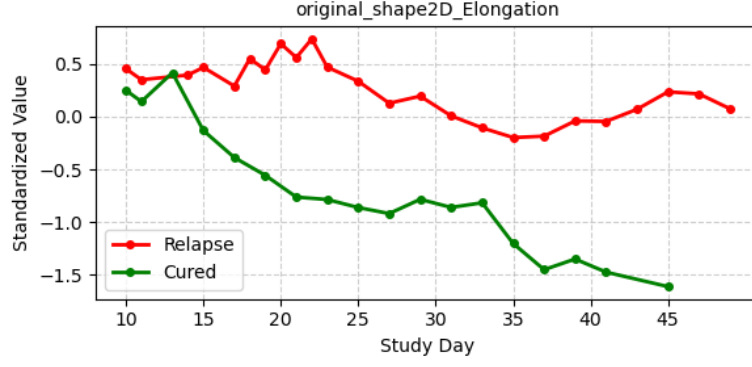


Figure 1: Elongation value evolution across cured and relapse groups.

Dependence Entropy (Figure 2) increased in cured animals during the later stages of treatment, while remaining comparatively stable in relapsing animals.

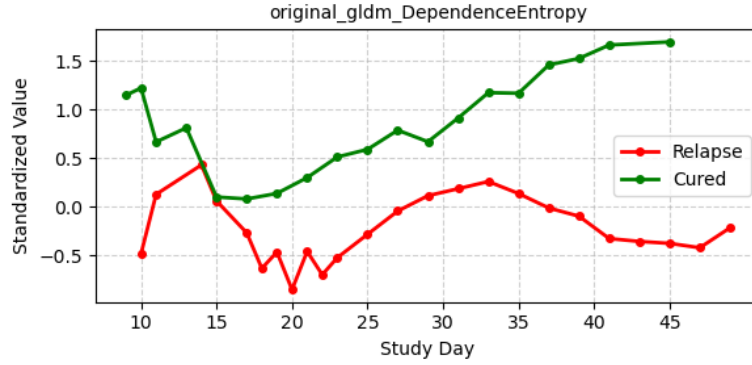


Figure 2: Dependence Entropy evolution across cured and relapse groups.

The repeated identification of these descriptors across the different feature-selection strategies supports their robustness within the present cohort. Additional low-dimensional visualizations of the selected radiomic feature space (t-SNE and UMAP projections) are provided in Supplementary Figure S2.

2.3 Deep Learning Model Analysis

Among all evaluated architectures and training configurations, the fine-tuned EfficientNetB0 model trained exclusively on grayscale MRI images achieved the best overall performance, with an AUC of 0.8511, an accuracy of 81.9%, and a sensitivity exceeding 90%.

Performance improved consistently along the transfer-learning axis. Training from scratch without mask information performed worst (AUC 0.5331); adding binary tumor masks raised discrimination substantially (AUC 0.7699); initializing from ImageNet weights provided a further boost, with the strongest frozen feature-extraction configuration reaching an AUC of 0.8583; and end-to-end fine-tuning produced the most balanced classifier (sensitivity 0.9091, NPV 0.9684). This ordering indicates that both pre-trained representations and full fine-tuning were the main drivers of performance.

The inclusion of tumor masks as additional input channels did not improve the fine-tuned model and occasionally reduced predictive accuracy. This result indicates that, given sufficient training capacity, the network was able to identify relevant regions directly from the MRI data without explicit localization guidance.

2.4 Temporal Prediction Performance

Predictive accuracy varied throughout tumor evolution. Both approaches exhibited lower performance during the earliest imaging time points, particularly before completion of the second TMZ administration, when tumor appearance remains relatively similar between future cured and future relapsing animals.

For the radiomics pipeline, misclassifications were concentrated in the earliest study days (approximately days 9–15), especially for cured animals, whose imaging signatures became distinctive only as therapeutic effects accumulated at later time points. Relapsing cases, on the contrary, were classified accurately and more uniformly across the timeline.

Despite this challenge, the DL model maintained substantially greater stability than the radiomics pipeline, markedly flattening the early-window error peak and improving per-subject consistency, with most animals surpassing 60% individual accuracy. This reduction of early-day vulnerability is particularly relevant clinically, as it enables more reliable prediction of treatment response at earlier stages. The per-day distribution of classification accuracy and misclassified examinations is reported in Supplementary Figure S3.

2.5 Interpretability and Error Analysis

Gradient-weighted Class Activation Mapping (Grad-CAM) visualizations showed that EfficientNet predictions were predominantly driven by anatomically meaningful regions (Figure 3). In correctly classified cases, attention maps consistently highlighted the tumor core and its immediate surroundings, and frequently extended into peritumoral regions.

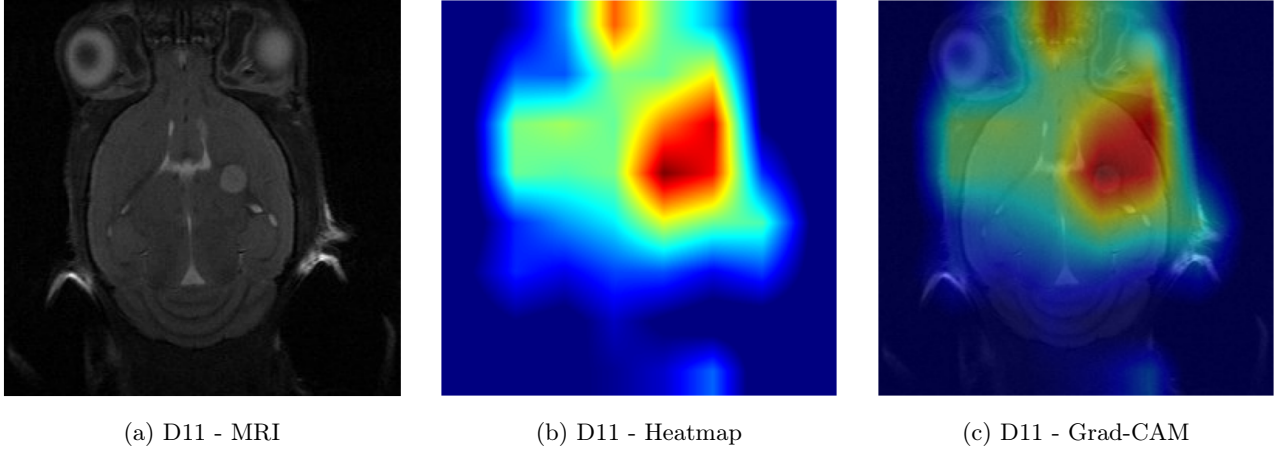


Figure 3: Day 11 visualization for subject C1263 (correctly predicted): original MRI, Grad-CAM heatmap, and Grad-CAM overlay. The temporal evolution of these maps across days 14, 17, and 19 is shown in Supplementary Figure S4.

Analysis of misclassified examinations revealed a different pattern: model attention frequently shifted toward anatomically irrelevant structures or became spatially diffuse. Tracking a representative relapsing subject (C1263) across consecutive time points illustrates this behavior: on the correctly classified Day 11 the attention was centered on the tumor, whereas on Day 14 the saliency map became diffuse and misaligned, on Day 17 it focused on the animal’s eye—an artifact-related region with no pathological significance—and on Day 19 it again failed to localize the tumor. Prediction failures were thus associated with uncertainty regarding lesion localization rather than an inability to learn discriminative tumor features. The evolution of Grad-CAM attention maps across these consecutive time points is provided in Supplementary Figure S4.

3 Discussion

This study investigated whether longitudinal T2-weighted MRI contains imaging signatures capable of predicting long-term therapeutic outcome in a preclinical GL261 glioblastoma model. Two complementary strategies were evaluated: a radiomics-based ML pipeline using handcrafted quantitative descriptors and an image-based deep learning framework built upon transfer learning. Although both approaches demonstrated predictive capability, the DL model consistently achieved superior performance across most evaluation metrics, suggesting that image-derived representations capture relevant information that is not fully represented by conventional radiomic features.

The radiomics approach successfully identified several interpretable imaging biomarkers whose temporal behavior differed between cured and relapsing tumors. Elongation reflects the geometric anisotropy of the lesion, and its progressive reduction in cured animals is consistent with the development of a more compact and structurally homogeneous tumor following successful therapy; in contrast, the persistently irregular morphology observed in relapsing tumors may indicate ongoing infiltrative growth. Texture-based descriptors such as Run Entropy and Dependence Entropy quantify the complexity and heterogeneity of spatial intensity distributions, where elevated values generally reflect greater local variability and structural disorder. The increase in Dependence Entropy observed in cured animals during later treatment stages may reflect tissue-remodeling processes, inflammatory responses, or treatment-induced structural reorganization occurring after tumor regression. Importantly, this trend, increasing entropy associated with a favorable outcome, highlights a potentially

distinctive biological signature of effective therapy in this preclinical model, which may be difficult to observe in clinical settings where complete cures are rare. These observations are consistent with previous radiomics studies demonstrating that texture heterogeneity carries prognostic information beyond conventional anatomical measurements, and they suggest that therapeutic response is accompanied by progressive microstructural changes detectable before substantial anatomical differences become apparent.

The superior performance of the DL model, particularly its high sensitivity (0.91) and its stability during early prediction windows, suggests that convolutional networks can capture more complex and subtle spatial patterns than handcrafted radiomic features. The success of transfer learning from a model pre-trained on natural images (ImageNet) underscores its value in overcoming data scarcity, a pervasive challenge in preclinical medical imaging: general-purpose low-level filters were effectively adapted to the specific task of identifying predictive patterns in T2-weighted MRI scans. Notably, the finding that explicit tumor masks did not improve performance, and could even degrade it, indicates that treatment-response signatures are likely distributed across both tumor and peritumoral tissue, rather than confined to the manually segmented lesion. This is coherent with the Grad-CAM analysis, in which attention frequently extended into peritumoral regions.

These results are aligned with, and extend, previous work on preclinical glioblastoma. Núñez et al. [16] and Ortega-Martorell et al. [17] demonstrated the value of radiomics and one-dimensional convolutional networks applied to multimodal MRI/MRSI data, while Zhao et al. [18] linked MR-based metabolomic signatures to immune-related mechanisms of relapse. Our findings indicate that a substantial fraction of the response-related information can already be recovered from standard T2-weighted MRI alone, without spectroscopic or multimodal acquisitions. In contrast to approaches that model radiomic dynamics as a jointly classified multivariate time series [9], we treated each time point as an individual observation while encoding temporal context through engineered descriptors and subject-aware validation; this design is better suited to the irregular and heterogeneous sampling of our cohort.

Although DL models are frequently criticized as “black boxes”, the Grad-CAM analysis demonstrates that their decisions can be interpreted and that correct predictions are consistently associated with anatomically relevant attention. Conversely, misclassifications were associated with diffuse or anatomically irrelevant attention, suggesting that errors stem primarily from uncertainty in lesion localization rather than from a failure to learn discriminative features. From a translational standpoint, the ability of the DL model to maintain performance during early prediction windows is particularly relevant, since the main clinical value of a predictive biomarker lies precisely in providing reliable information before conventional radiological assessment becomes conclusive.

Several limitations should be considered when interpreting these results. First, the study was conducted using a relatively small cohort of 18 animals, which restricts statistical power and may limit generalizability. Although Grouped Leave-One-Out validation was employed to minimize information leakage and provide a realistic estimate of model performance, larger independent cohorts will be necessary to confirm the robustness of the identified biomarkers.

Second, all analyses were performed using a single preclinical glioblastoma model (GL261) and data acquired at a single institution. While the GL261 model reproduces several important biological characteristics of human glioblastoma, additional validation in alternative experimental models and clinical datasets will be required before translational conclusions can be drawn.

Third, the radiomics pipeline relied on manual tumor segmentations, introducing potential observer-dependent variability. Automated or semi-automated segmentation approaches may improve reproducibility in future studies.

Finally, only T2-weighted MRI was analyzed. The incorporation of additional imaging modalities, including diffusion MRI, perfusion imaging, spectroscopy, or multimodal radiomic signatures, may provide complementary information and further improve predictive performance.

4 Conclusion

We have demonstrated that both radiomics and DL are viable strategies for predicting therapy outcomes in preclinical glioblastoma from longitudinal MRI. However, a DL approach based on fine-tuning a pre-trained CNN outperforms the classical radiomics pipeline in terms of accuracy, sensitivity, and temporal stability. The DL model’s ability to identify early and subtle markers of treatment response, combined with methods for interpreting its decisions, makes it a highly promising tool for developing non-invasive biomarkers. This work lays a strong foundation for future research aimed at creating more reliable and personalized methods for monitoring glioblastoma treatment.

Future work should focus on external validation, multimodal integration, and longitudinal modeling. In particular, combining conventional MRI with spectroscopic, metabolic, molecular, or histopathological information may provide a more comprehensive representation of tumor evolution. Furthermore, temporal DL architectures capable of explicitly modeling sequential imaging examinations could better exploit the longitudinal nature of therapeutic response. Such developments may ultimately contribute to clinically deployable systems for early

treatment adaptation and personalized glioblastoma management.

5 Materials and Methods

An overview of the two analytical pipelines evaluated in this study is presented in Figure 4. Their details are provided in the following subsections.

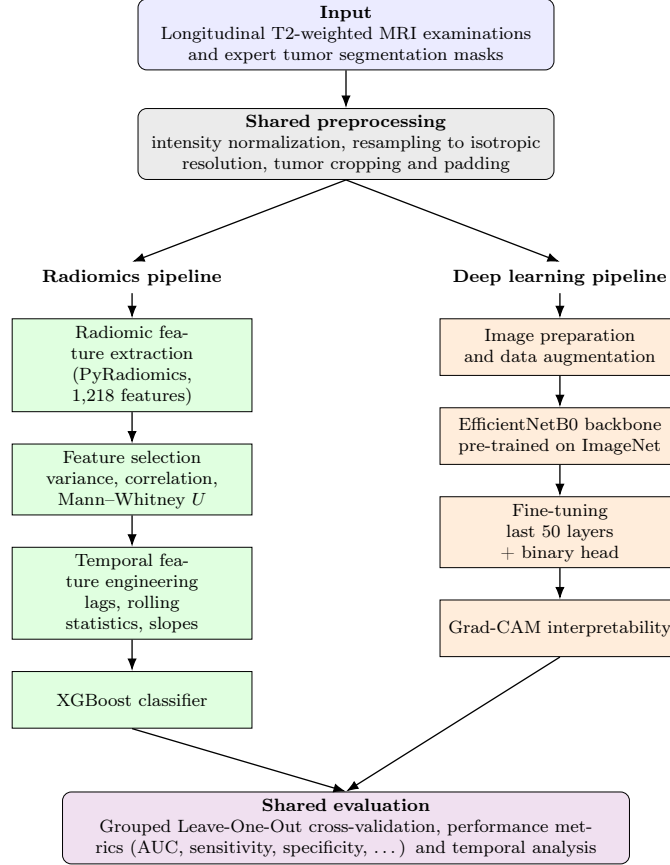


Figure 4: Overview of the experimental workflow. Longitudinal T2-weighted MRI examinations and their corresponding tumor masks share a common preprocessing stage and are then processed through two parallel pipelines: a radiomics-based pipeline coupled with an XGBoost classifier and a deep learning pipeline based on a fine-tuned EfficientNetB0 model. Both pipelines are evaluated under the same Grouped Leave-One-Out cross-validation scheme.

5.1 Preclinical Dataset

This study used a retrospective dataset of longitudinal T2-weighted (T2w) MRI scans acquired from 18 C57BL/6 mice orthotopically implanted with GL261 GB cells and monitored throughout tumor evolution [4]. These tumor-bearing mice were either control/untreated or treated with TMZ following an immune-enhancing metronomic schedule [19]. The animals were divided into three outcome groups:

- **Control (n=8):** untreated mice, showing natural tumor progression.
- **Relapsing (n=5):** mice treated with TMZ that showed a transient response followed by tumor regrowth.
- **Cured (n=5):** mice treated with TMZ that showed complete tumor regression and long-term survival.

The dataset is longitudinal in nature: each animal underwent a variable number of imaging examinations acquired at different time points along tumor evolution, so that the number of explorations and the range of days studied differ across subjects (Figure 5). A summary of the cohort is presented in Table 2, and a detailed per-subject description (number of images per exploration, number of explorations, and range of days studied) is provided in Supplementary Table S1.

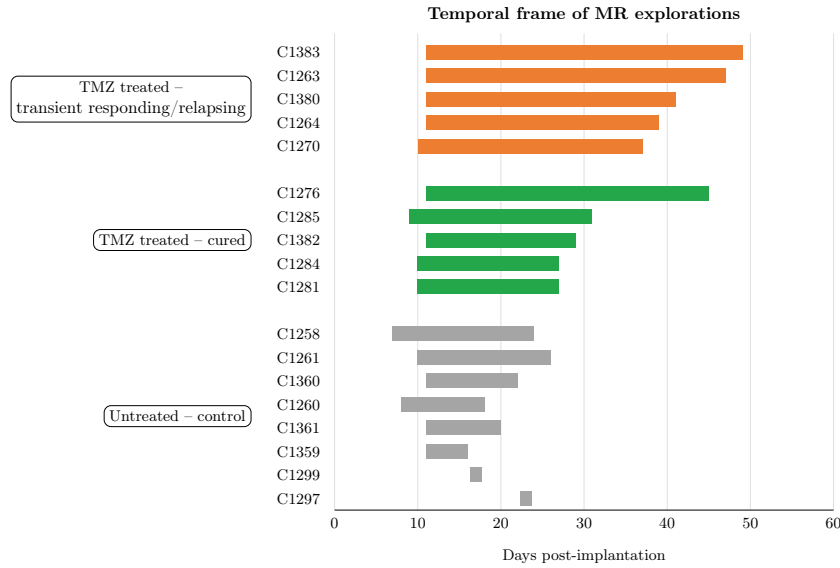


Figure 5: Temporal frame of the MRI explorations for each animal in the cohort. Each horizontal bar spans the range of days post-implantation over which a given subject (CXXXX) was imaged, grouped by experimental condition: TMZ-treated transient responding/relapsing (orange), TMZ-treated cured (green), and untreated control (grey). Day 0 corresponds to the tumor inoculation day; subjects C1297 and C1299 underwent a single exploration. Per-subject details are reported in Supplementary Table S1.

The animal experiments from which these data were obtained were approved by the Ethics Committee on Animal Experimentation (CERec) of the Universitat Autònoma de Barcelona (protocols CEA-OH-9685 and CEEAH-3665), and conducted in accordance with the 3Rs principles (Replacement, Reduction, and Refinement), Spanish legislation (Real Decreto 53/2013), and European Directive 2010/63/EU on the protection of animals used for scientific purposes.

Imaging was performed on a 7T Bruker BioSpec 70/30 USR spectrometer (Bruker BioSpin GmbH, Ettlingen, Germany), equipped with a mini-imaging gradient set (400 mT/m). A 72-mm inner-diameter linear volume coil was used as a transmitter, and a dedicated mouse brain quadrature surface coil was used as a receiver. Mice were positioned in a bed, which allowed delivery of anaesthesia (isoflurane, 1.5–2.0% in O₂ at 1 L/min), with an integrated heating water circuit for body temperature regulation. Respiratory frequency was monitored with a pressure probe and kept between 60–80 breaths/min. GB-bearing mice were screened with high-resolution coronal T2w MRI using a Rapid Acquisition with Relaxation Enhancement (RARE) sequence to evaluate the presence of brain tumors and monitor their stage of evolution, with repetition time (TR)/effective echo time (TE_{eff}) = 4200/36 ms. For each MRI scan, tumor regions were manually segmented by experts to create binary masks. The core analysis focused on discriminating between the "Cured" and "Relapsing" groups to specifically model treatment-response dynamics. **Animals were monitored daily by trained veterinary personnel according to a standardized welfare scoring system aimed at assessing animal well-being and detecting signs of tumour progression. The scoring system evaluated several clinical and behavioural criteria. A body weight loss exceeding 20% of the initial weight was considered a predefined humane endpoint and led to euthanasia. Animals were then euthanized by cervical dislocation under isoflurane anaesthesia, as described above.**

A representative example of the longitudinal acquisition is shown in Figure 6, which illustrates consecutive T2-weighted examinations of a single treated animal together with the corresponding expert tumor segmentations across several time points.

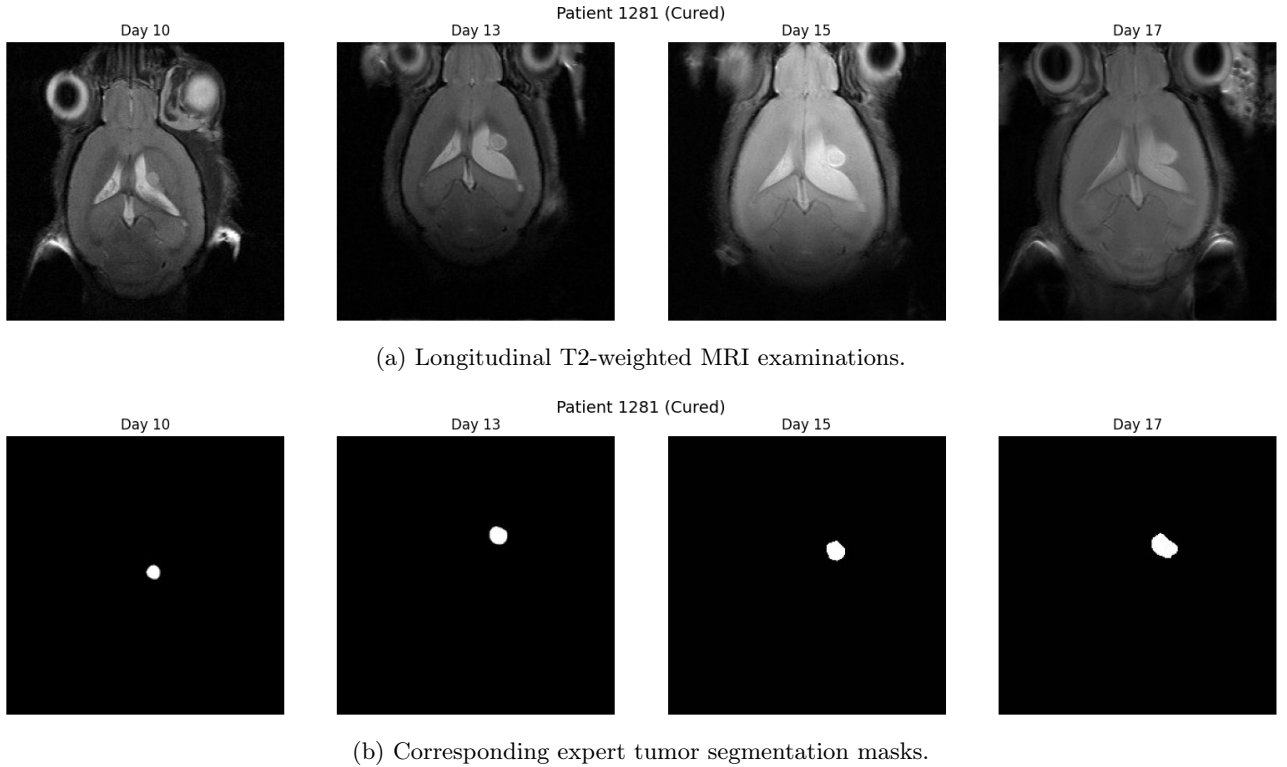


Figure 6: Illustration of the longitudinal nature of the dataset for a representative treated (cured) animal (subject C1281). (a) Consecutive T2-weighted MRI examinations acquired at different days along tumor evolution and treatment, and (b) their corresponding manually delineated binary tumor masks. Each animal contributes several such examinations acquired at irregular time points, which are treated as individual observations rather than as a jointly modeled temporal sequence.

Group	Mice	MRI Examinations	Outcome
Control	8	36	Natural progression
Relapsing	5	84	Initial response followed by recurrence
Cured	5	49	Complete response
Total	18	169	—

Table 2: Summary of the experimental cohort. MRI examinations correspond to longitudinal imaging sessions acquired throughout tumor evolution. A detailed per-subject breakdown is available in Supplementary Table S1.

5.2 Study Design and Longitudinal Data Handling

Although the dataset is longitudinal, complete temporal sequences were not modeled jointly as a multivariate time series [9]. Instead, each imaging time point was treated as an individual labeled observation. Temporal information was incorporated in two complementary ways: (i) in the radiomics pipeline, through engineered temporal descriptors computed across consecutive examinations of the same animal (see below); and (ii) in the evaluation, through a subject-aware cross-validation scheme and a temporal analysis of predictive performance. This design reflects the heterogeneous and irregular sampling of the cohort, in which the number and timing of examinations vary substantially across subjects.

5.3 Radiomics-Based Pipeline

The radiomics workflow was designed to transform longitudinal MRI examinations into a structured set of quantitative descriptors capable of capturing tumor morphology, intensity characteristics, and intratumoral heterogeneity throughout treatment. Given the limited sample size and high dimensionality of the feature space, particular emphasis was placed on robust preprocessing, dimensionality reduction, and model regularization to minimize overfitting and maximize generalization.

5.3.1 Image Preprocessing

Prior to feature extraction, all MRI scans underwent a standardized preprocessing procedure implemented according to recommendations from the Image Biomarker Standardization Initiative (IBSI) [20]. Intensity normalization was performed to reduce acquisition-related variability and improve reproducibility across imaging sessions. A fixed gray-level discretization bin width of 1 was applied to ensure consistency in texture calculations. To comply with PyRadiomics [11] requirements and avoid negative intensity values, a voxel array shift of +1000 was introduced. Extreme intensity values corresponding to potential imaging artifacts were trimmed using a three-standard-deviation threshold. To ensure spatial consistency between subjects and acquisition sessions, all images were resampled to an isotropic in-plane resolution of 0.2 mm using B-spline interpolation. Tumor masks were corrected when necessary and padded with a five-voxel margin to preserve relevant contextual information surrounding the lesion. Finally, images were cropped around the tumor region to reduce computational requirements while maintaining biologically relevant tissue information.

5.3.2 Radiomic Feature Extraction

Radiomic features were extracted using the PyRadiomics framework from manually segmented tumor regions on T2-weighted MRI slices. A total of 1,218 quantitative features were computed for each image, including first-order statistics, shape-based descriptors, and second- and higher-order texture features (GLCM, GLRLM, GLSZM, GLDM, and NGTDM). Feature extraction was performed not only on the original MRI images but also on multiple transformed representations, including Laplacian-of-Gaussian (LoG) filtered images at multiple scales ($\sigma = 1.0, 2.0, \text{ and } 3.0$), one-level wavelet decompositions, gradient images, and intensity transformations (square, square-root, logarithmic, and exponential). A complete description of the feature families and image transformations is provided in Supplementary Methods S4.

5.3.3 Feature Selection and Dimensionality Reduction

Given the extremely high dimensionality of the radiomic feature space relative to the number of available subjects, a rigorous multi-stage feature selection strategy was implemented. First, missing values were imputed using mean-value substitution, and features exhibiting near-zero variance (variance < 0.01) were removed. Redundancy was then reduced through correlation filtering using Spearman rank correlation coefficients: when two features exhibited a correlation magnitude greater than 0.95, one of them was discarded. Subsequently, statistical filtering was performed using the Mann-Whitney U test, retaining only features with statistically significant differences between cured and relapsing mice ($p < 0.05$); this non-parametric test was selected because most radiomic variables deviated substantially from normality. Finally, all selected features were standardized using z-score normalization.

5.3.4 Temporal Feature Engineering

Because tumor evolution is inherently dynamic, static radiomic measurements may fail to capture biologically meaningful treatment trajectories. To incorporate longitudinal information into the predictive framework, temporal descriptors were generated from the most relevant radiomic features. For each selected feature, lag-based variables were computed to characterize temporal evolution across consecutive imaging sessions of the same animal, including first-order temporal differences, rolling means, rolling standard deviations, local temporal slopes, and relative percentage changes. Such descriptors capture not only the current tumor phenotype but also the direction and rate of change of imaging biomarkers throughout treatment. Importantly, this strategy encodes temporal context without treating the data as a jointly modeled multivariate time series.

5.3.5 Classification Using XGBoost

The final classification stage was performed using eXtreme Gradient Boosting (XGBoost) [21], an ensemble learning algorithm based on gradient-boosted decision trees. XGBoost was selected because it is particularly well-suited to radiomics applications characterized by structured tabular data, limited sample sizes, high-dimensional feature spaces, and complex non-linear interactions among variables. Its built-in L1 and L2 regularization, shrinkage, column subsampling, and tree pruning mechanisms reduce overfitting risk in small datasets, while feature importance scores enable biological interpretation of the resulting models.

Hyperparameter optimization was conducted through grid search embedded within the training procedure. The final configuration is summarized in Supplementary Table S3, and the complete search grid is reported in Supplementary Table S2. To address class imbalance between cured and relapsing mice, several resampling strategies were investigated, including the Synthetic Minority Over-sampling Technique (SMOTE) [22], Edited Nearest Neighbors (ENN) [23], and training on the original class distribution. Comparative evaluation demonstrated that controlled oversampling improved sensitivity for the minority cured class while preserving overall model discrimination.

5.4 Deep Learning-Based Pipeline

While radiomics relies on handcrafted descriptors, DL enables the automatic extraction of hierarchical feature representations directly from imaging data. This capability is particularly relevant in glioblastoma, where treatment response may be encoded through complex spatial patterns that cannot be fully captured by manually engineered features. Because the application of DL in preclinical imaging is constrained by the limited number of available subjects, a transfer learning strategy was adopted to reduce the risk of overfitting. Additional architectural details are provided in Supplementary Methods S1.

5.4.1 Image Preparation and Preprocessing

Each MRI examination consisted of a T2-weighted image together with its corresponding manually delineated tumor mask. Prior to network training, images were resized to the input resolution required by the network while preserving anatomical proportions, and intensity values were normalized to match the distribution expected by ImageNet-pretrained models. Data augmentation (random rotations, horizontal and vertical translations, zoom operations, and intensity perturbations) was applied during training to increase dataset diversity and improve generalization. Control cases were excluded from training, as untreated animals do not reflect the therapeutic dynamics under study. To handle the dataset efficiently, all samples were converted to the TFRecord format and processed through a TensorFlow `tf.data` pipeline enabling on-the-fly augmentation, shuffling, batching, and prefetching.

5.4.2 EfficientNetB0 Architecture and Transfer Learning

The backbone architecture was EfficientNetB0, a CNN proposed by Tan and Le [24] that employs a compound scaling strategy jointly optimizing network depth, width, and input resolution. EfficientNetB0 was selected because it provides competitive accuracy with only approximately 5.3 million trainable parameters, substantially fewer than architectures such as VGG16, ResNet50, or DenseNet121, which is advantageous in small medical imaging datasets. The network was initialized with ImageNet weights [25]. Initially, the convolutional backbone was frozen, allowing only the classification head to learn task-specific representations; subsequently, the upper layers were unfrozen and fine-tuned with a reduced learning rate, adapting the model to glioblastoma MRI characteristics while preserving robust low-level representations.

5.4.3 Fine-Tuning Procedure

Fine-tuning replaced the original ImageNet classification head with a custom binary classification head discriminating between cured and relapsing mice. The last 50 layers of the backbone were unfrozen, the model accepted three-channel input images, and the classification head consisted of a global pooling layer, a dense layer, batch normalization, dropout, and a final sigmoid activation. Training used a binary cross-entropy loss and the Adam optimizer, with a batch size of 32 and random oversampling of the minority class applied only to the training partition within each fold to avoid information leakage. Dropout and early stopping were used for regularization, and the checkpoint corresponding to the best validation performance was retained. The full fine-tuning configuration is summarized in Supplementary Table S4.

5.4.4 Input Configurations

To investigate the contribution of explicit tumor localization information, two input configurations were evaluated. In the first, the network received only the grayscale T2-weighted MRI image, allowing it to autonomously identify relevant anatomical structures. In the second, manually generated tumor masks were incorporated as additional image channels, explicitly guiding the network toward the lesion. Comparing both configurations enabled assessment of whether prior anatomical knowledge improves predictive performance or whether the network can independently identify the most informative regions.

5.5 Evaluation and Interpretability

5.5.1 Performance Metrics

Model performance was assessed using a comprehensive set of complementary metrics. The primary metric was the AUC, which measures discrimination between cured and relapsing mice independently of any specific decision threshold. In addition, Accuracy, Sensitivity (recall of the cured class), Specificity, Positive Predictive Value (PPV), Negative Predictive Value (NPV), and F1-score were computed. Because the cured and relapsing groups were not perfectly balanced, sensitivity was considered of special interest, as failing to identify a truly responding subject may have important implications for treatment assessment and biomarker discovery.

5.5.2 Validation Strategy

A GLOO cross-validation framework was employed for both pipelines. In longitudinal imaging studies, multiple MRI examinations are acquired from the same animal, so observations from the same mouse exhibit substantial intra-subject correlation and cannot be considered statistically independent; random train-test splitting would therefore introduce severe information leakage. To avoid this, all MRI examinations corresponding to a given mouse were assigned exclusively to either the training or testing set within each fold, and one complete animal was withheld for testing at every iteration. Performance values were aggregated across all folds to obtain robust estimates and reduce dependence on any individual animal.

5.5.3 Temporal Evaluation

Beyond global performance, predictive behavior was analyzed as a function of tumor evolution time, since treatment-induced imaging changes emerge progressively after therapy administration. Particular attention was devoted to the period between treatment initiation and the onset of visible tumor regression, which represents the most clinically relevant prediction window, as conventional radiological assessment often remains inconclusive during early treatment stages.

5.5.4 Interpretability Analysis

Interpretability analyses were conducted for both paradigms. For the radiomics pipeline, feature importance values extracted from the XGBoost classifier were averaged across cross-validation folds to obtain a robust ranking of predictive biomarkers, and the temporal evolution of the most influential features was analyzed to characterize differences between cured and relapsing tumors. For the DL framework, Grad-CAM [26] was used to generate class-specific attention maps for both correctly classified and misclassified cases, enabling investigation of whether network decisions were based on biologically meaningful tumor regions. Additional methodological details of Grad-CAM are provided in Supplementary Methods S3.

Data availability

The dataset analyzed during the current study was sourced from previously published experiments and is available from the corresponding author upon reasonable request.

Acknowledgements

The authors thank the joint preclinical MRI facility of the Universitat Autònoma de Barcelona and the Centro de Investigación Biomédica en Red-Bioingeniería, Biomateriales y Nanomedicina (CIBER-BBN), where all MRI experiments were carried out, as part of Unit 25 of NANBIOSIS (Cerdanyola del Vallès, Spain).

Author contributions

J.G.: methodology, software, formal analysis, investigation, data curation, visualization, and writing of the original draft.

A.P.C.: conceptualization, data acquisition, supervision, funding acquisition, and manuscript review and editing.

A.V.: conceptualization, methodology, supervision, funding acquisition, and manuscript review and editing.

Funding

This work was funded by MICIU/AEI/10.13039/501100011033/ (grants PID2023-147750NB-I00 to A.P.C. and PID2022-143299OB-I00 to A.V.), Generalitat de Catalunya (2021 SGR 0135), and Instituto de Salud Carlos III (CIBER-BBN group CB06-01-0010).

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information accompanies this manuscript, comprising Supplementary Methods, Supplementary Tables S1–S4, and Supplementary Figures S1–S5.

References

- [1] R. Stupp, W. P. Mason, M. J. van den Bent, M. Weller, B. Fisher, M. J. Taphoorn, et al. Radiotherapy plus concomitant and adjuvant temozolomide for glioblastoma. *N. Engl. J. Med.*, 352(10):987–996, 2005.
- [2] G. Shukla, G. S. Alexander, S. Bakas, R. Nikam, K. Talekar, J. D. Palmer, and W. Shi. Advanced magnetic resonance imaging in glioblastoma: a review. *Chin. Clin. Oncol.*, 2017.
- [3] P. Liu, S. Griffiths, D. Veljanoski, P. Vaughn-Beaucaire, V. Speirs, and A. Brüning-Richardson. Preclinical models of glioblastoma: limitations of current models and the promise of new developments. *Expert Rev. Mol. Med.*, 23:e20, 2021.
- [4] S. Wu, P. Calero-Pérez, L. Vilamañan, N. Arias-Ramos, M. Pumarola, S. Ortega-Martorell, et al. Anti-tumour immune response in GL261 glioblastoma generated by temozolomide immune-enhancing metronomic schedule monitored with MRSI-based nosological images. *NMR Biomed.*, 33(4):e4229, 2020.
- [5] V. Kumar, Y. Gu, S. Basu, A. Berglund, S. A. Eschrich, M. B. Schabath, et al. Radiomics: the process and the challenges. *Magn. Reson. Imaging*, 30:1234–1248, 2012.
- [6] A. Hosny, C. Parmar, J. Quackenbush, L. H. Schwartz, and H. J. W. L. Aerts. Artificial intelligence in radiology. *Nat. Rev. Cancer*, 18(8):500–510, 2018.
- [7] K. A. Tran, O. Kondrashova, A. Bradley, E. D. Williams, J. V. Pearson, and N. Waddell. Deep learning in cancer diagnosis, prognosis and treatment selection. *Genome Med.*, 13:152, 2021.
- [8] D. Shen, G. Wu, and H.-I. Suk. Deep learning in medical image analysis. *Annu. Rev. Biomed. Eng.*, 19:221–248, 2017.
- [9] F. Prinzi, A. Orlando, S. Gaglio, and S. Vitabile. Breast cancer classification through multivariate radiomic time series analysis in DCE-MRI sequences. *Expert Syst. Appl.*, 249:123557, 2024.
- [10] C. Pitarch, G. Ugan, M. Julià-Sapé, and A. Vellido. Advances in the use of deep learning for the analysis of magnetic resonance image in neuro-oncology. *Cancers*, 16(2):300, 2024.
- [11] J. J. M. van Griethuysen, A. Fedorov, C. Parmar, N. Aucoin, V. Narayan, R. G. H. Beets-Tan, et al. Computational radiomics system to decode the radiographic phenotype. *Cancer Res.*, 77(21):e104–e107, 2017.
- [12] E. I. Styliara, L. G. Astrakas, G. Alexiou, V. G. Xydis, A. Zikou, G. Kafritsas, et al. Survival outcome prediction in glioblastoma: insights from MRI radiomics. *Curr. Oncol.*, 31(4):2233–2243, 2024.
- [13] F. J. Dorfner, J. B. Patel, J. Kalpathy-Cramer, E. R. Gerstner, and C. P. Bridge. A review of deep learning for brain tumor analysis in MRI. *npj Precis. Oncol.*, 9:1–13, 2025.
- [14] M. Bonada, L. F. Rossi, G. Carone, F. Panico, F. Cofano, P. Fiaschi, et al. Deep learning for MRI segmentation and molecular subtyping in glioblastoma: critical aspects from an emerging field. *Biomedicines*, 12(8):1878, 2024.
- [15] A. Çinar and M. Yildirim. Detection of tumors on brain MRI images using the hybrid convolutional neural network architecture. *Med. Hypotheses*, 139:109684, 2020.
- [16] L. M. Núñez, M. Julià-Sapé, E. Romero, C. Arús, A. Vellido, and A. P. Candiota. Interpreting response to TMZ therapy in murine GL261 glioblastoma by combining radiomics, convex-NMF and feature selection in MRI/MRSI data analysis. In *2020 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8, 2020.
- [17] S. Ortega-Martorell, I. Olier, O. Hernandez, P. D. Restrepo-Galvis, R. A. A. Bellfield, and A. P. Candiota. Tracking therapy response in glioblastoma using 1D convolutional neural networks. *Cancers*, 15(15):4002, 2023.

- [18] K. Zhao, P. Calero-Pérez, M. Mulero-Acevedo, M. H. A. Bopp, V. Möschl, A. Pagenstecher, et al. Correlation of MR-based metabolomics and molecular profiling in the tumor microenvironment of temozolomide-treated orthotopic GL261 glioblastoma in mice. *Int. J. Mol. Sci.*, 24(24):17628, 2023.
- [19] J. Wu and D. J. Waxman. Metronomic cyclophosphamide eradicates large implanted GL261 gliomas by activating antitumor CD8⁺ T-cell responses and immune memory. *Oncoimmunology*, 4(4):e1005521, 2015.
- [20] A. Zwanenburg, M. Vallières, M. A. Abdalah, H. J. W. L. Aerts, V. Andrearczyk, A. Apte, et al. The image biomarker standardization initiative: standardized quantitative radiomics for high-throughput image-based phenotyping. *Radiology*, 295(2):328–338, 2020.
- [21] T. Chen and C. Guestrin. XGBoost: a scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.
- [22] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer. SMOTE: synthetic minority over-sampling technique. *J. Artif. Intell. Res.*, 16:321–357, 2002.
- [23] D. L. Wilson. Asymptotic properties of nearest neighbor rules using edited data. *IEEE Trans. Syst. Man Cybern.*, SMC-2(3):408–421, 1972.
- [24] M. Tan and Q. V. Le. EfficientNet: rethinking model scaling for convolutional neural networks. In *Proceedings of the 36th International Conference on Machine Learning*, pages 6105–6114, 2019.
- [25] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei. ImageNet: a large-scale hierarchical image database. In *2009 IEEE Conference on Computer Vision and Pattern Recognition*, pages 248–255, 2009.
- [26] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra. Grad-CAM: visual explanations from deep networks via gradient-based localization. *Int. J. Comput. Vis.*, 127(2):313–338, 2019.